

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Empathy Map Canvas

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

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Empathy Map Canvas

1. Introduction

This document presents the Empathy Map Canvas developed for the primary stakeholders of the Online Complaint Registration System [cite: 1074]. Empathy mapping is a design-thinking technique used to understand users' behaviors, emotions, motivations, challenges, and expectations [cite: 1075].

The purpose of this activity is to gain deeper insight into how different stakeholders experience the complaint management process and identify opportunities for improving service delivery through technology [cite: 1075].

Two key stakeholders have been identified: [cite: 1076]

- The Concerned Citizen (Complaint User)
- The Service Resolution Agent

Each persona is analyzed across six dimensions: Think & Feel, Hear, See, Say & Do, Pains, and Gains [cite: 1077]. This analysis ensures that the design and implementation decisions remain aligned with actual user needs rather than assumptions [cite: 1075].

2. Persona 1 – "The Concerned Citizen" [cite: 1078]

Name: The Concerned Citizen [cite: 1080]

Profile: A citizen between 18–60 years of age who regularly interacts with public services and wishes to report civic issues such as electricity problems, water supply disruptions, sanitation concerns, road damages, and other public grievances [cite: 1081]. The citizen expects transparency, quick responses, and visibility throughout the complaint resolution process [cite: 1082].

2.1 Empathy Map – Quadrant Chart

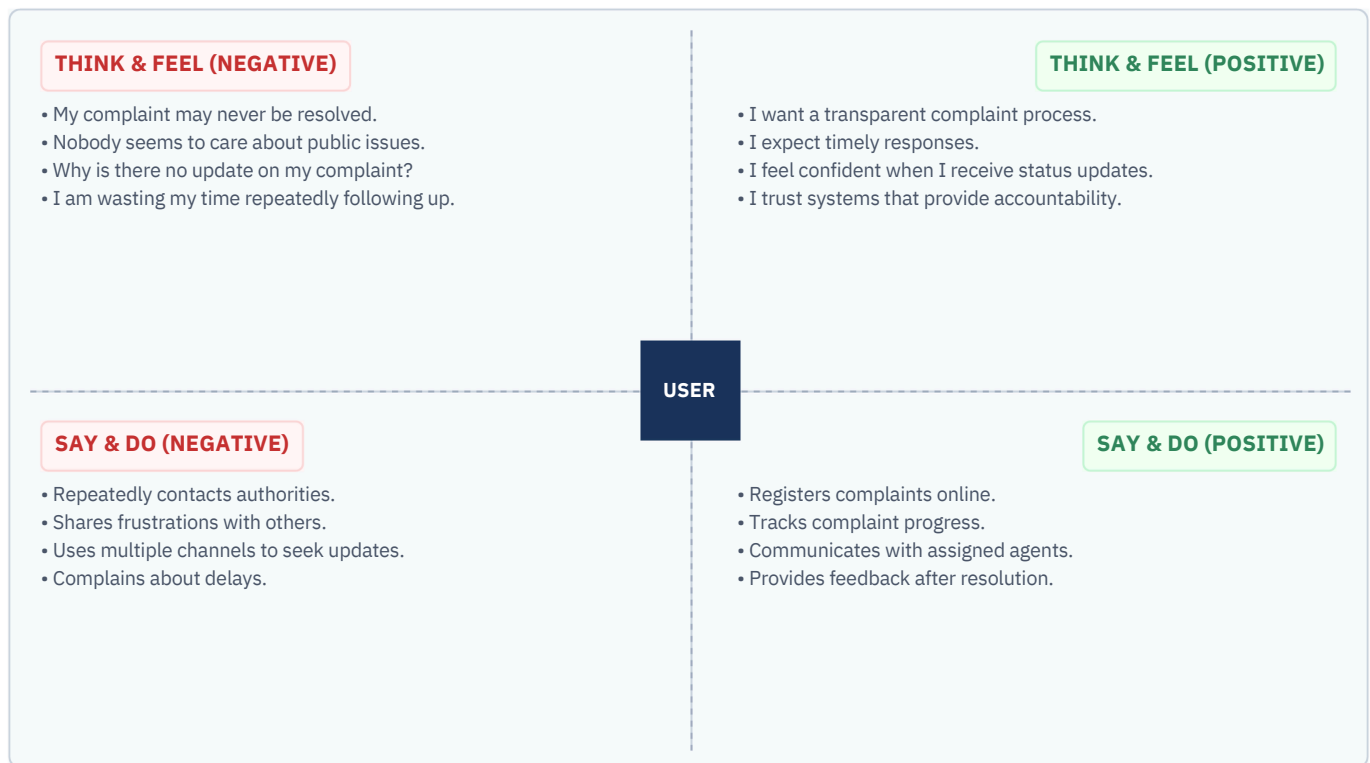


FIGURE 1.1 – EMPATHY MAP QUADRANT CHART: PERSONA 1 – THE CONCERNED CITIZEN

2.2 Detailed Dimension Analysis

THINK & FEEL [cite: 1104]

- "I hope someone actually reviews my complaint."
- "I want to know exactly what stage my complaint is currently in."
- Feels anxious when complaints remain unresolved for long periods.
- Feels reassured when the system provides regular updates and clear status tracking.
- Believes transparency increases trust in service providers.

HEAR [cite: 1110]

- "You should register it online; it's easier to track."
- "The authorities never respond on time."
- "My complaint got resolved quickly after using the portal."
- Friends, family members, and community members influence expectations regarding complaint resolution services [cite: 1116].

SEE [cite: 1117]

- Traditional complaint systems requiring physical visits.

- Long waiting times for issue resolution.
- Social media complaints from other citizens.
- Modern government services offering digital tracking and notifications.

SAY & DO [cite: 1118]

- "I need to know when my issue will be fixed."
- "Can I directly communicate with the assigned technician?"
- Registers complaints through the portal.
- Tracks complaint status frequently.
- Uses real-time chat to seek updates.
- Submits feedback after resolution.

PAINS [cite: 1130]

- Lack of transparency in traditional systems.
- Delayed responses from authorities.
- No direct communication channel.
- Difficulty tracking complaint progress.
- Repeated follow-ups required.
- Uncertainty about resolution timelines.

GAINS [cite: 1136]

- Real-time complaint tracking.
- Direct communication with agents.
- Instant notifications.
- Transparency throughout complaint lifecycle.
- Faster complaint resolution.
- Confidence in service delivery.

3. Persona 2 – "The Service Resolution Agent" [cite: 1141]

Name: Service Resolution Agent [cite: 1144]

Profile: A government or service department employee responsible for resolving citizen complaints [cite: 1145, 1146]. The agent manages assigned complaints, updates statuses, communicates with citizens, and ensures timely issue resolution [cite: 1146]. The agent values efficiency, centralized information, and streamlined workflows [cite: 1146].

3.1 Detailed Dimension Analysis [cite: 1147]

THINK & FEEL [cite: 1148]

- "I need all complaint information in one place."
- "I want to resolve complaints without unnecessary delays."
- Feels pressured when handling multiple complaints simultaneously [cite: 1151].
- Wants better communication with citizens.
- Needs clear assignment and prioritization mechanisms.

HEAR [cite: 1155]

- "Citizens are waiting for updates."
- "Resolution times need improvement."
- "The department wants better service quality metrics."
- Management emphasizes accountability and efficiency.

SEE [cite: 1159]

- Large numbers of incoming complaints.
- Different complaint categories requiring specialized handling.
- Manual systems causing inefficiencies.
- Growing expectations for digital services.

SAY & DO [cite: 1164]

- "Let me check the complaint details."
- "I will update the status once the issue is resolved."
- Reviews assigned complaints.
- Updates complaint progress.
- Uses chat functionality to communicate with citizens.
- Maintains service records.

PAINS [cite: 1169]

- Manual complaint management [cite: 1170].
- Lack of centralized communication [cite: 1170].
- Difficulty prioritizing complaints [cite: 1172].
- No real-time collaboration tools.
- Administrative overhead.
- Delayed information flow.

GAINS [cite: 1173]

- Centralized complaint dashboard [cite: 1174].
- Real-time communication system.
- Efficient assignment workflow.
- Quick status updates [cite: 1178].
- Improved productivity.
- Better citizen satisfaction.

4. Persona Comparison Summary [cite: 1179]

Dimension	Persona 1 – Concerned Citizen	Persona 2 – Service Resolution Agent
Primary Goal	Register and track complaints [cite: 1180]	Resolve assigned complaints efficiently [cite: 1180]
Core Pain	Lack of transparency and delayed updates [cite: 1180]	Manual workflows and communication challenges [cite: 1180]
Core Gain	Real-time updates and complaint visibility [cite: 1180]	Centralized management and faster resolution [cite: 1180]
Key Requirement	Complaint tracking and notifications [cite: 1180]	Assignment management and status updates [cite: 1180]
Communication Need	Direct interaction with agents [cite: 1180]	Efficient citizen communication
Technology Support	User dashboard and chat system [cite: 1180]	Agent dashboard and complaint workspace [cite: 1180]
Success Metric	Faster issue resolution	Improved operational efficiency

5. Conclusion [cite: 1181]

The empathy maps reveal the expectations, frustrations, and motivations of the two primary stakeholders of the Online Complaint Registration System [cite: 1182]. The Concerned Citizen seeks transparency, accountability, and efficient issue resolution, while the Service Resolution Agent requires streamlined workflows, centralized information, and effective communication tools [cite: 1183].

The proposed MERN-based platform addresses these requirements through core design components [cite: 1184]: Complaint Registration, Complaint Tracking, Role-Based Access Control, Agent Assignment Workflow, Real-Time Chat using Socket.IO, Notification System, Feedback Collection, and an Analytics Dashboard. Every feature of the system directly contributes toward reducing identified user pains and enhancing user gains, thereby improving both citizen satisfaction and operational efficiency [cite: 1185].